SIA: Structure Invariant Attack
Transfer Attack on Face Verification

Aditi Raj (IGDTUW)

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Selecting SIA

Why SIA?

- ICCV 2023 — core ML/vision conference
- Official implementation in TransferAttack repository
- Works with CNN-based models (no ViT required)
- Not implemented in intern baseline
- Input-transformation attack: strongest category for transfer

Paper

Wang et al., “Structure Invariant Transformation for better Adversarial Transferability”, ICCV 2023. arXiv:2309.14700

Code

https://github.com/aditiraj2006/transferattacktInterns

Understanding SIA

SIA = Structure Invariant Attack — input-transformation transfer attack

Problem with existing attacks

- TI-FGSM, SIM apply one transform to the whole image
- All copies look similar
- Limited diversity $\Rightarrow$ perturbation overfits to surrogate
- Moderate transfer only

Momentum Update (same as MI-FGSM)

- Maintains accumulated gradient
- g = decay*g + grad_avg

SIA:Block-level Diversity

- Split image into K $\times$ K blocks
- Apply a different random transform to each block indep endently
- 10^{K2} possibleimagevariants per iteration
- Average gradient over N = 20 diverse copies

Attack Objectives (unchanged)

- Impersonation: maximize cosine similarity
- Dodging: minimize cosine

Structure Invariant Transformation (SIT)

How the block-level transformation works:

Step1:Split face into blocks

- Divide image into 4 ×4=16 blocks
- Spatial structure preserved: eyes left/right, nose center, mouth b elow
- Identity-relevant layout intact

Step 2: Transform each block randomly

- Each block independently chooses from 10 transforms
- 10^{16} possible image variants

Step 3: Average gradient over N=20copies

10 Block-levelTransforms:

0. Identity
1. Horizontal flip
2. Scale down (80%)
3. Grayscale
4. Contrast enhance (×1.3)
5. Brightness increase (+0.1)
6. Vertical flip
7. Rotation 90 ◦
8. Rotation 180 ◦
9. Rotation 270 ◦

Each block independently chooses one of these 10 transforms at each iteration

Code Integration

Step 1 — Register SIA

```
ALL_ATTACKS= [
  PGD, MI_FGSM,
  TI_FGSM, SI_NI_FGSM,
  MI_ADMIX_DI_TI ,
  SIA  # <-- added
]
```

Step 3 — Dispatcher

```
if  attack_name == "SIA":
      return  sia_attack(...)
```

Step 2 — sia_attack() **Function**

1. Initialize adversarial image
2. Initialize momentum g = 0
3. For T iterations:
4. For N = 20 copies:
5. Apply SIT (block transforms)
6. Compute gradient
7. Average N gradients
8. Normalize gradient (L1)
9. Update momentum
10. Update adversarial image
11. Project into ϵ-ball
12. Clip to valid range

Adapting SIA to Face Verification

Original SIA (ImageNet classification)

- Target: fool class label prediction
- Loss: cross-entropy on logits
- PyTorch, standard CNN classifiers
- Single classification loss per image

Adaptation required

- Face verification uses cosine similarity in embedding space, not class logits
- TensorFlow/DeepFace models (notPyTorch)

What was preserved (unchanged)

- 4×4 block-splitting grid
- All 10 transform types
- N = 20 copies per iteration
- MI-FGSM momentum update
- L-inf ϵ-ball projection

What was adapted

- Loss: cosine similarity (impersonation: maximize; dodging: minimize)
- Gradient via `tf.GradientTape`
- _preprocess() for per-model input sizes

Key Challenges Encountered

Block Rotation Shape Mismatch

- Rotating a non-square block changes dimensions: $14 \times 18 \rightarrow 18 \times 14$ after 90°
- Fixed: explicit resize after each block transform to restore original block shape

Dataset Path Mismatch

- Subset CSV paths had absolute prefixes
- Fixed: `fix_path()` strips any known prefix and joins with `faces_dir`

Model Input Size Mismatch

- ArcFace, GhostFaceNet: 112×112
- Facenet512: 160×160
- VGG-Face: 224×224
- Fixed: per-model resizing in `_preprocess()`

Computational Cost

- N = 20 gradient calls per step — ~4× slower than MI-FGSM
- Used N = 3 for quick tests, N = 20 for final evaluation
- Total runtime: ~3 hours on CPU for full subset

Final Results

Attack	Breach Rate (%)	Goal
SIA (proposed)	**34.17**	overall
impersonation	45.83	–
dodging	22.50	–
SI-NI-FGSM	31.46	(best baseline)
MI-FGSM	26.04	
MI-ADMIX-DI-TI	23.96	
TI-FGSM	20.00	
PGD	16.46	

SIA outperformed ALL baselines:

- +2.71% over previous best (SI-NI-FGSM)
- +8.13% over MI-FGSM
- +17.71% over PGD
- Strong impersonation: 45.83% breach rate

What I Learned

Technical

- How input-transformation diversity directly improves transfer
- Why block-level transforms outperform global transforms
- Adapting a classification attack to face verification (cosine similarity)
- Per-model input preprocessing in DeepFace
- Handling variable block shapes after rotation
- TF GradientTape for custom gradient computation

Process

- Reading and understanding a full research paper
- Integrating a new attack into an existing codebase without breaking anything
- Quantitative comparison: breach rate vs. embedding impact
- Debugging silent failures (path mismatches, shape errors)
- Choosing an attack that is both novel and fits the setting
- Extending baselines without redesigning the system

Thank You

Aditi Raj — IGDTUW

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